Labsheet -04

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```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df = pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_and_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-selected-columns.csv")

print(df.head())
print(df.shape)
```

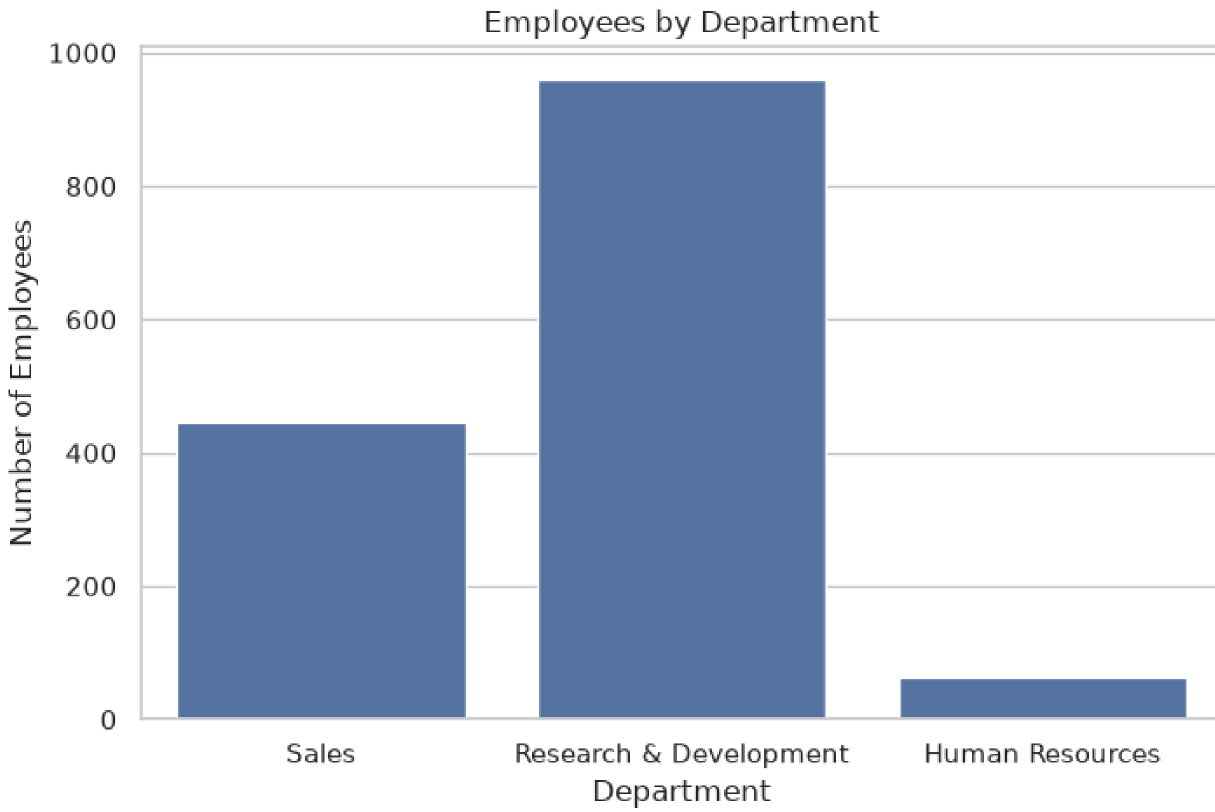
```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7
(1470, 10)
```

```python
print(df.columns.tolist())
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber']
```

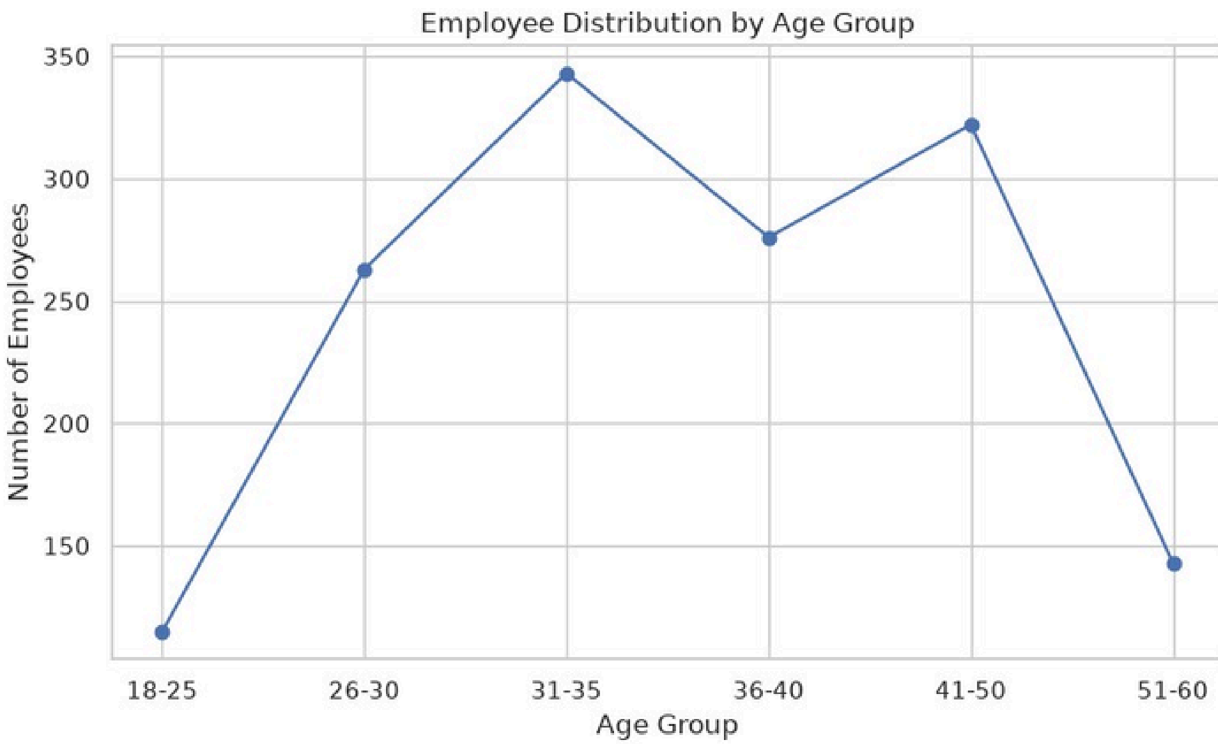
```python
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x="Department")

plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")

plt.show()
```



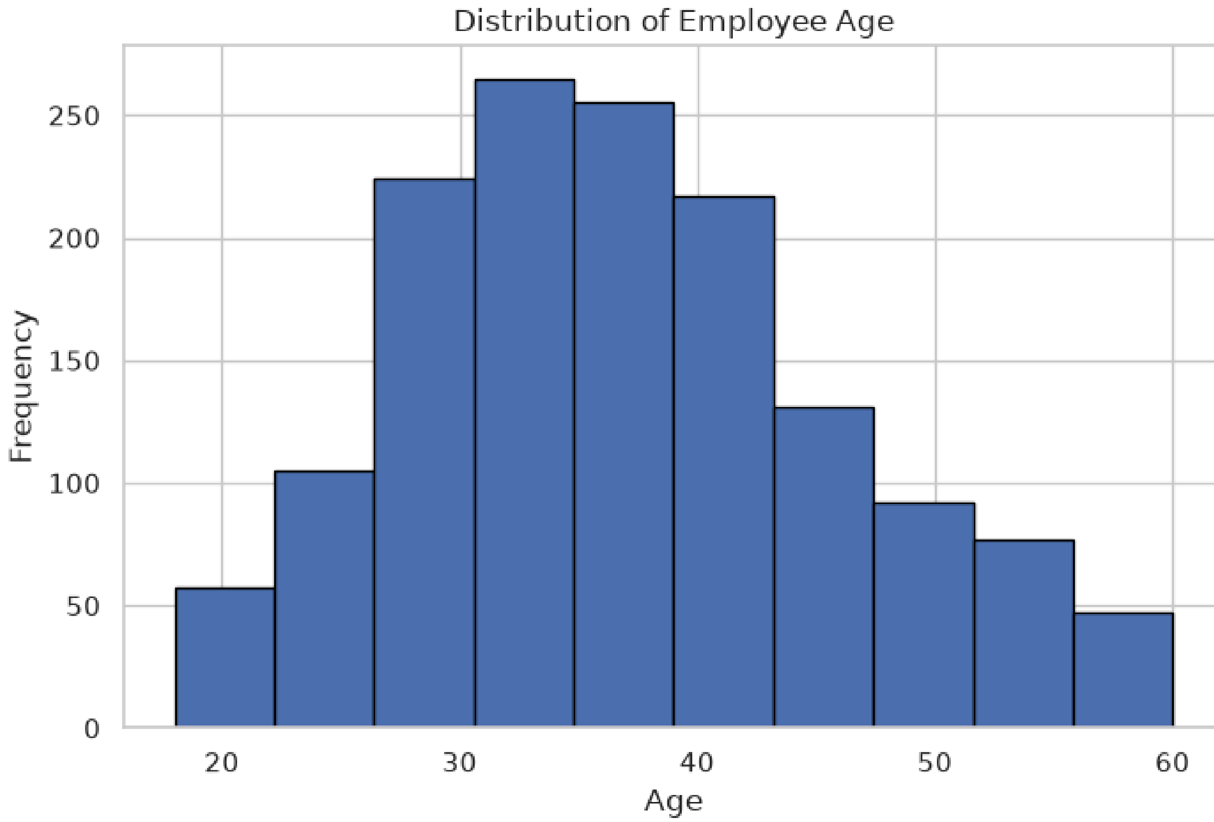

```
df["AgeGroup"] = pd.cut(
        df["Age"],
        bins=[18, 25, 30, 35, 40, 50, 60],
        labels=["18-25", "26-30", "31-35", "36-40", "41-50", "51-60"]
)

age_count          =          df["AgeGroup"].value_counts().sort_index()
plt.figure(figsize=(9, 5))

plt.plot(age_count.index,          age_count.values,          marker="o")
plt.title("Employee   Distribution   by   Age   Group")   plt.xlabel("Age
Group")
plt.ylabel("Number of Employees")
```

```
plt.show()
```

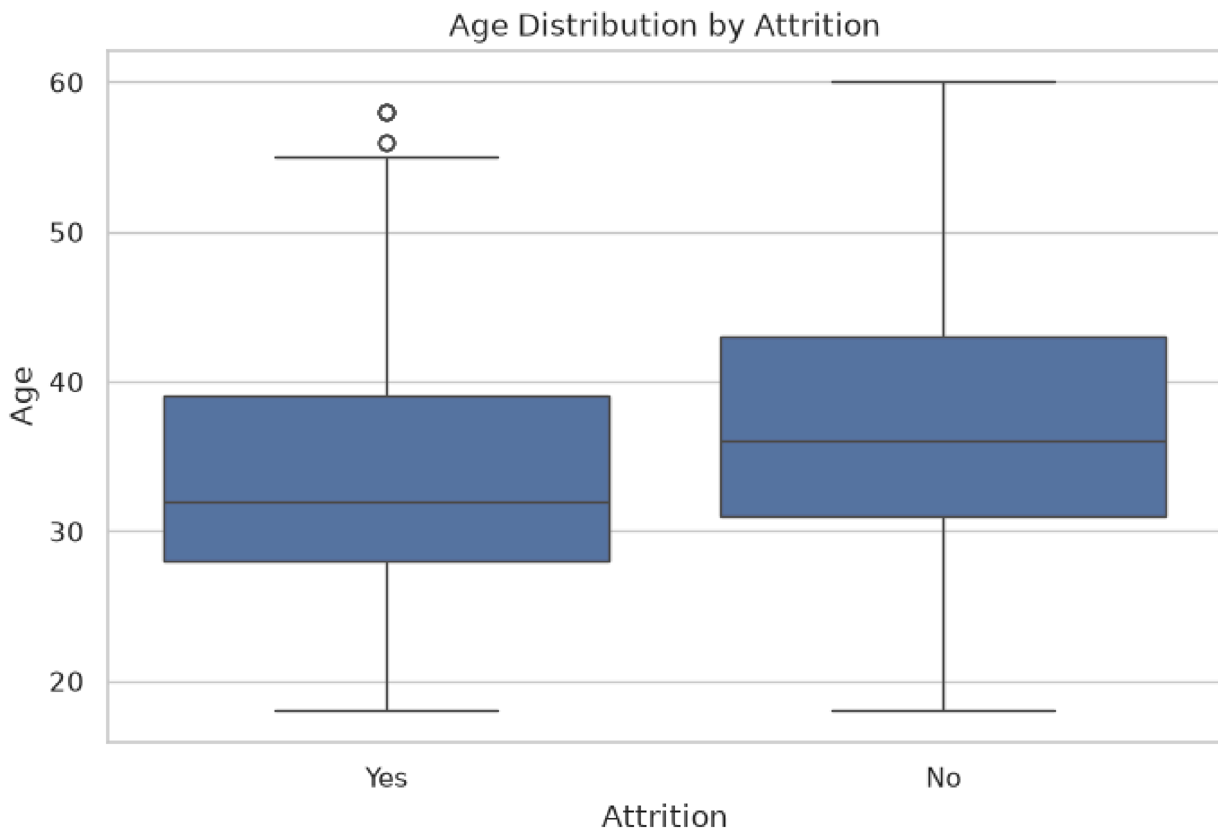



```
plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=10, edgecolor="black")

plt.title("Distribution of Employee Age")

plt.xlabel("Age")

plt.ylabel("Frequency")
```

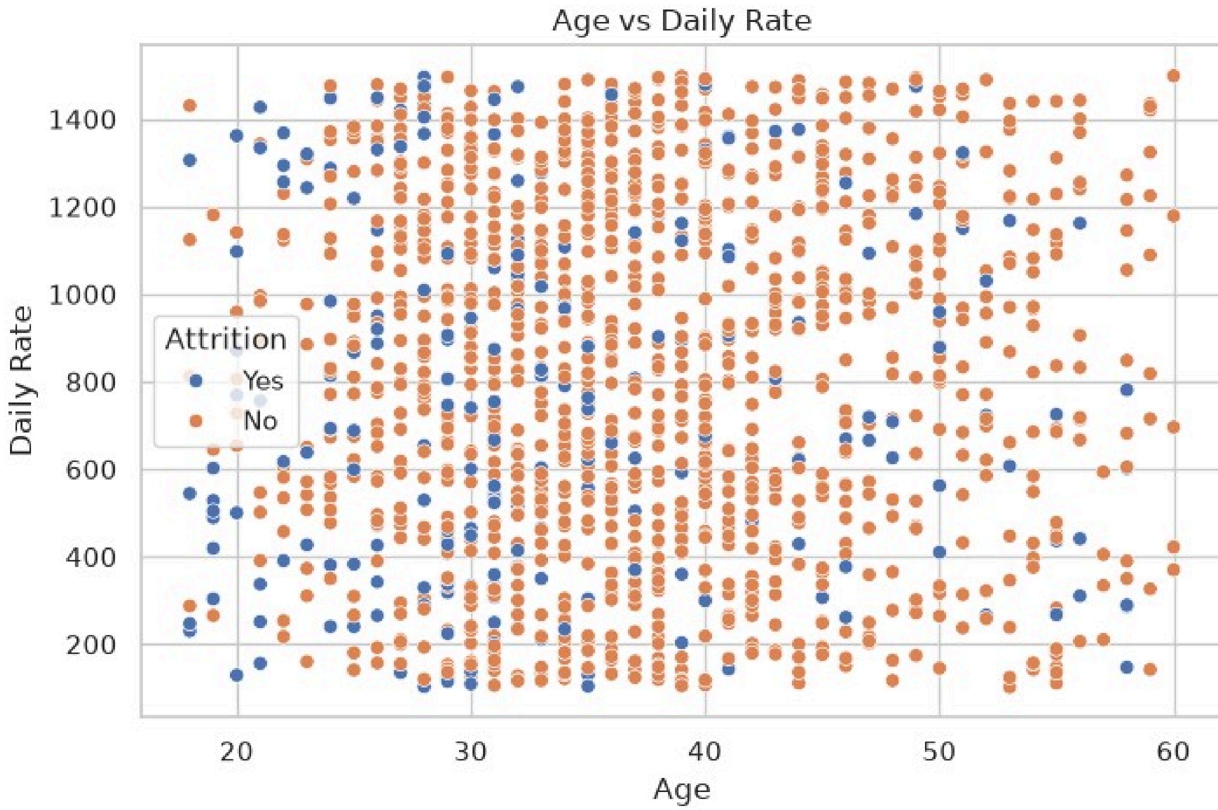
```
plt.show()
```



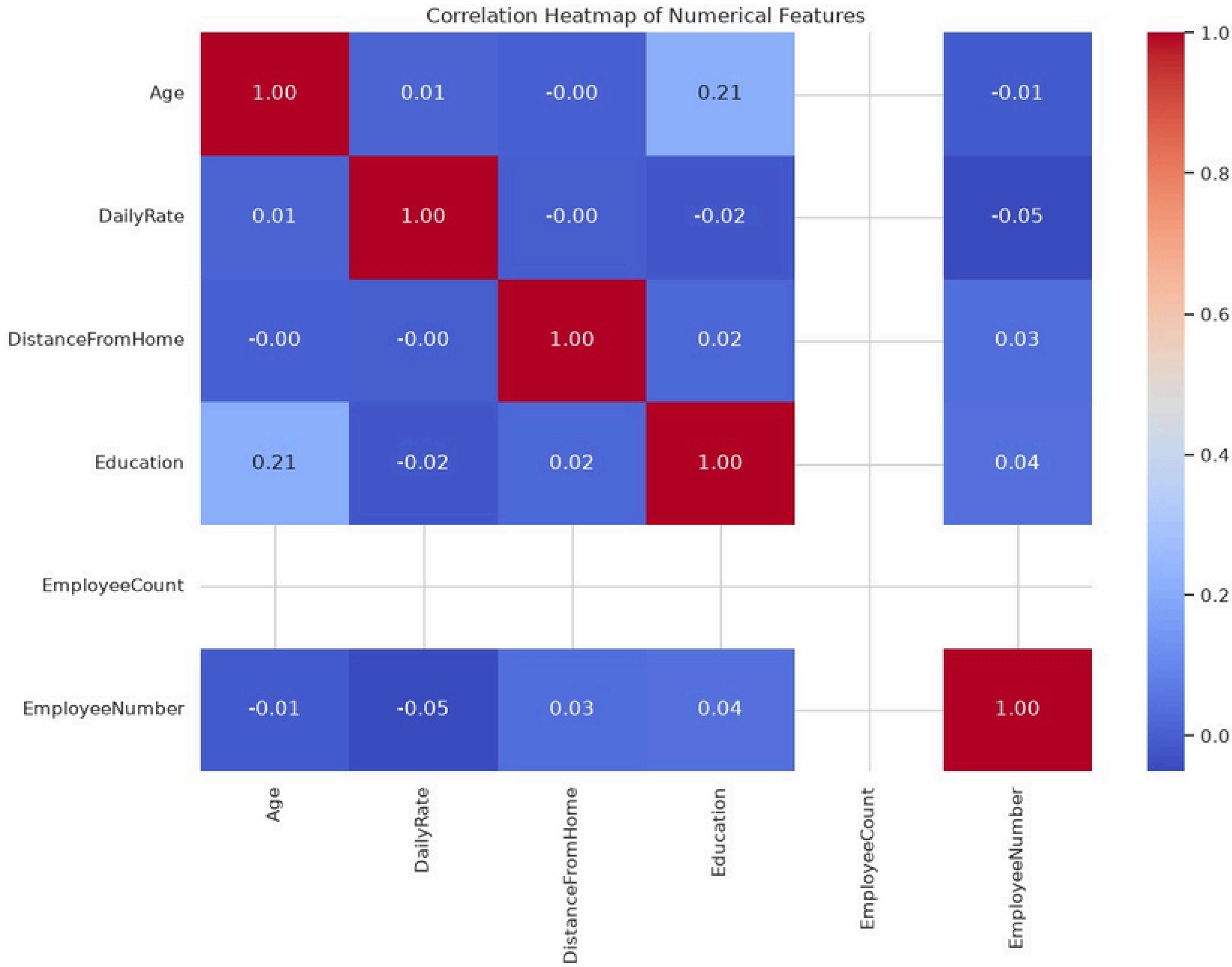

```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Attrition", y="Age")

plt.title("Age Distribution by Attrition")

plt.xlabel("Attrition")

plt.ylabel("Age")

plt.show()
```



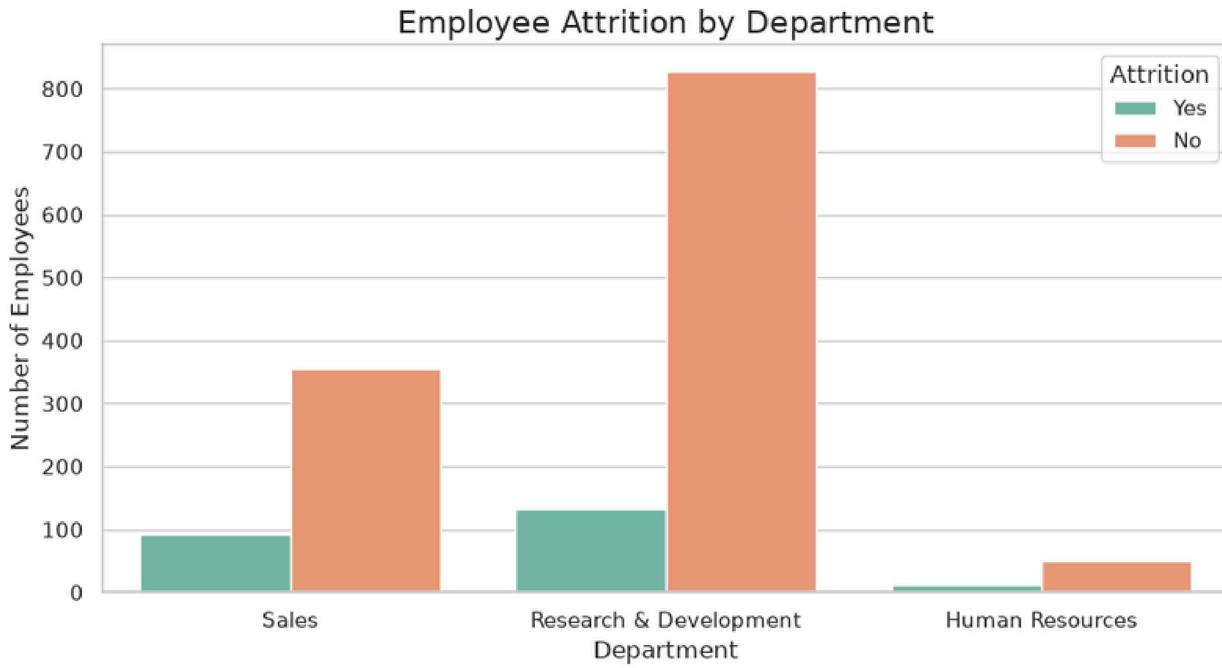

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
        data=df,
        x="Age",
        y="DailyRate",
        hue="Attrition"
)

plt.title("Age      vs     Daily      Rate")
plt.xlabel("Age")       plt.ylabel("Daily
Rate")

plt.show()
```




```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(12, 8))

sns.heatmap(
       numeric_df.corr(),
       annot=True,
       cmap="coolwarm",
       fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features") plt.show()
```




```
plt.figure(figsize=(9, 5))

sns.countplot(
    data=df,
    x="Department",
    hue="Attrition",
    palette="Set2"
)

plt.title("Employee Attrition by Department", fontsize=16)
plt.xlabel("Department", fontsize=12)
plt.ylabel("Number of Employees", fontsize=12)

plt.legend(title="Attrition")

plt.tight_layout()
plt.show()
```




```python
print("Total Employees:", len(df))

print("\nDepartment-wise Employees:")
print(df["Department"].value_counts())

print("\nAttrition Distribution:")
print(df["Attrition"].value_counts())

print("\nAverage Age:", df["Age"].mean())
print("\nAverage Daily Rate:", df["DailyRate"].mean())
```

```
Total Employees: 1470

Department-wise Employees:
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64

Attrition Distribution:
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Average Age: 36.923809523809524
Average Daily Rate: 802.4857142857143
```